Poor 31/100

Modelling and Control of Wheel Shimmy

Coursework for EE4-47 Modelling and Control of Multibody Mechanical System

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1. Introduction

Caster wheel shimmy is the self-excited oscillation that happen to common equipment such as automobile, trollies, wheelchairs etc. It is very important to understand wheel shimmy in such mechanical systems in order to increase their stability. Practical cases are extremely complex so in the scope of this report, a much simpler model is constructed and analysed to demonstrate the control of wheel shimmy, shown in figure 1. (Evangelou, n.d.)

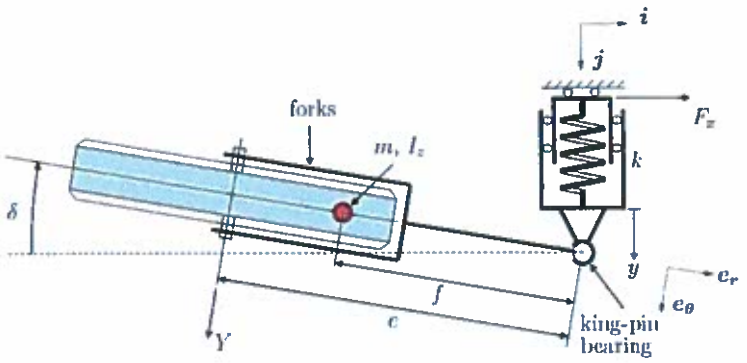


Figure 1. Top view of a simple system capable of shimmy

There are several assumptions related to this model. The wheel is axisymmetric and free to spin relative to the forks that support it; the wheel has no spin inertia. The king-pin is free to translate laterally with displacement y from static equilibrium and is assumed massless so that analysis deals with only one body.

2. Modelling, analysis and control tasks

Question 1

1.1 Cartesian System

- Position Vector:

$$\underline{r} = x\boldsymbol{i} + y\boldsymbol{j} + z\boldsymbol{k}$$

- Forces on mass:

$$F = mg\boldsymbol{k} + (F_x \cos\delta)\boldsymbol{i} + (F_x \sin\delta)\boldsymbol{j}$$

- Acceleration Vector:

$$m\frac{\partial^2 \underline{r}}{\partial t^2} = m\ddot{x}\boldsymbol{i} + m\ddot{y}\boldsymbol{j} + m\ddot{z}\boldsymbol{k} = F$$

- $\therefore m\ddot{x} = F_x \cos\delta$
  $m\ddot{y} = F_x \sin\delta$
  $m\ddot{z} = mg$
  $\therefore m\frac{\partial^2 \underline{r}}{\partial t^2} = F_x \cos\delta\, \boldsymbol{i} + F_x \sin\delta\, \boldsymbol{j} + mg\boldsymbol{k}$
  $\therefore \frac{\partial^2 \underline{x}}{\partial t^2} = \frac{F_x \cos\delta}{m} \qquad x = \frac{F_x \cos\delta}{2m}t^2 + C_1 t$
  $\therefore \frac{\partial^2 \underline{y}}{\partial t^2} = \frac{F_x \sin\delta}{m} \qquad x = \frac{F_x \sin\delta}{2m}t^2 + C_2 t$

1.2 Polar System

- Position Vector:

$$\underline{f} = f\boldsymbol{e_r}$$

$$\underline{e} = e\boldsymbol{e_r}$$

- Velocity Vector:

$$\dot{\underline{f}} = \frac{df}{dt} = \frac{d}{dt}(f\boldsymbol{e_r}) = \dot{f}\boldsymbol{e_r} + f\frac{d\boldsymbol{e_r}}{dt} = \dot{f}\boldsymbol{e_r} + f\frac{d\boldsymbol{e_r}}{d\theta}\frac{d\theta}{dt} = 0 + f\dot{\theta}\boldsymbol{e_\theta}$$

$$\dot{\underline{e}} = \frac{de}{dt} = \frac{d}{dt}(e\boldsymbol{e_r}) = \dot{e}\boldsymbol{e_r} + e\frac{d\boldsymbol{e_r}}{dt} = \dot{e}\boldsymbol{e_r} + e\frac{d\boldsymbol{e_r}}{d\theta}\frac{d\theta}{dt} = 0 + e\dot{\theta}\boldsymbol{e_\theta}$$

- Acceleration Vector:

$$\ddot{\underline{f}} = -f\dot{\theta}^2\boldsymbol{e_r} + f\ddot{\theta}\boldsymbol{e_\theta}$$

$$\ddot{\underline{e}} = -e\dot{\theta}^2\boldsymbol{e_r} + e\ddot{\theta}\boldsymbol{e_\theta}$$

- Forces:

$$F = m\ddot{\underline{f}} = F_1\boldsymbol{e_r} + F_2\boldsymbol{e_\theta} + mg\boldsymbol{k}$$

Question 2

Question 3

Subject to nonholonomic constrains, the velocity of the wheel is proportional to the angle of rotation about its axis.

incomplete

Question 4

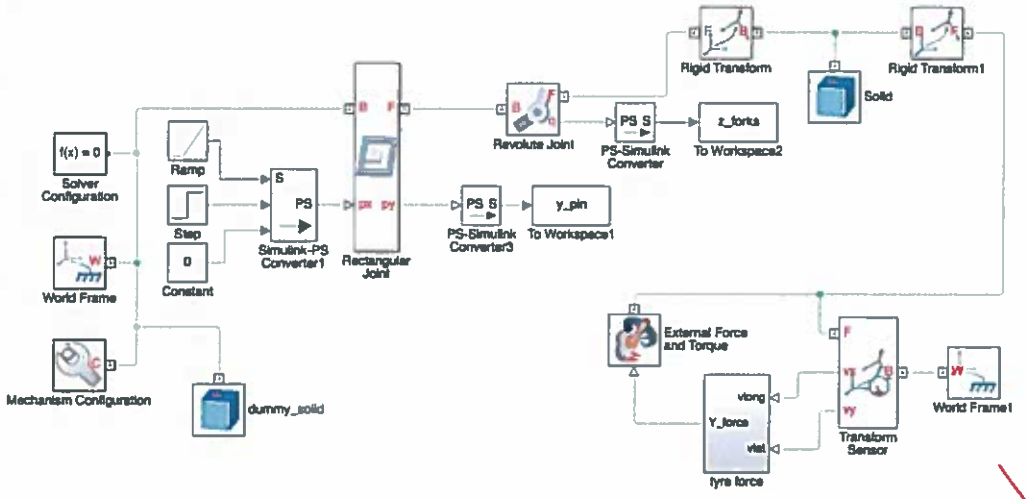


Figure 2. Simmechanics Model

Question 5

e = 0.7 + 0.3 = 1.0 m

f = 0.7 m

m = 200 kg

Iz = 150 kgm^2

C = 40000

Summary of Comments on Microsoft Word - Modelling and Control of Wheel Shimmy.docx

Page: 3

Number: 1 Author: Simos Evangelou Date: Indeterminate
Simos Evangelou: Not attempted.

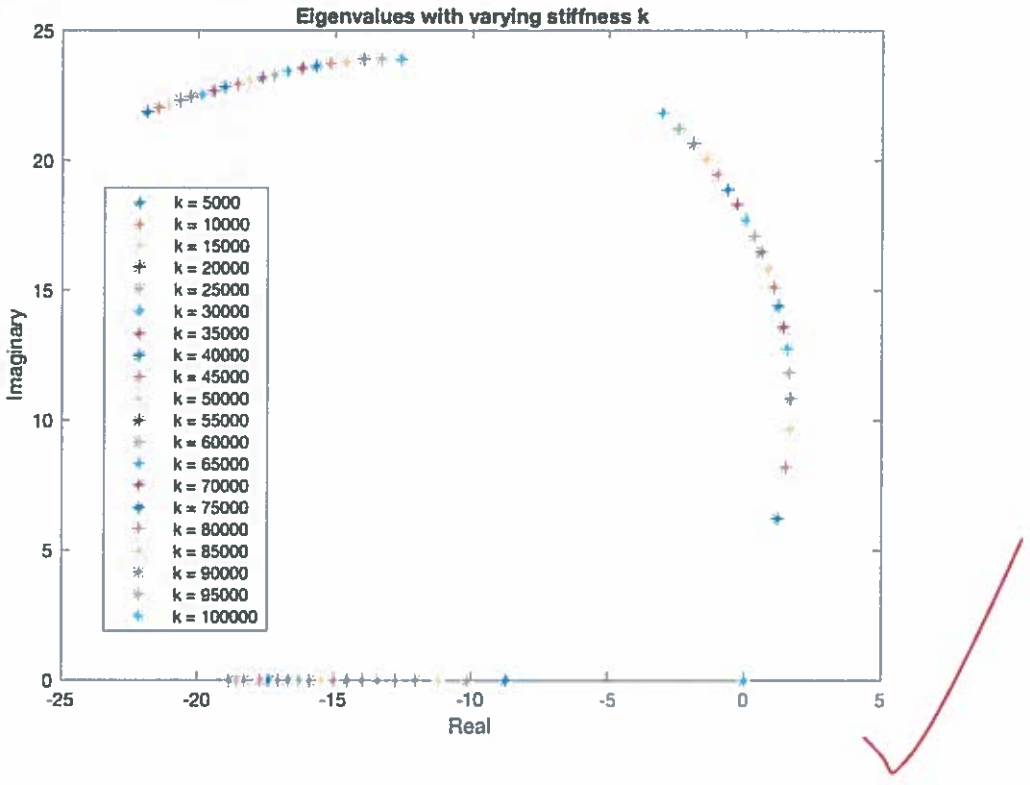
Number: 2 Author: Simos Evangelou Date: Indeterminate
Simos Evangelou: f = 0.6 m

Question 6

- Script

```
clear all
clc
speed=10;
z_initial=0;
sigma_rel=0.2;
for i=1:20
    k=5000+5000*(i-1);
    [A,B,C,D]=linmod('coursework_model');
    e=eig(A)
    figure(1)
    plot(real(e),imag(e),'*') % Plot real and imaginary parts
    xlabel('Real')
    ylabel('Imaginary')
    title('Eigenvalues with varying stiffness k');
    legendInfo{i}=['k = ' num2str(k)];
    i=i+1;
    hold on
end
legend(legendInfo,'southwest')
ylim([0,25])
hold off
```

- Plot



- There are three sets of eligible eigenvalues. In two of the three sets, eigenvalues are positive, suggesting the contribution does not decay to a fixed point, so they exhibit instability. As stiffness increases, eigenvalue becomes more and more positive, so the system becomes more and more instable.

Number: 1 Author: Simos Evangelou Date: Indeterminate

Simos Evangelou: No. As stiffness increases they become a bit more unstable but then they become less unstable and eventually stable for high stiffness values.

Question 7

- Script

```
%Parameters
speed=10;
z_initial=0.01;
sigma_rel=0.2;

%Simulation
k=30000;
A1=sim('coursework_model','StopTime','4');
y1=A1.get('y_pin');
z1=A1.get('z_forks');
figure
plot(z1,y1)
hold off
```



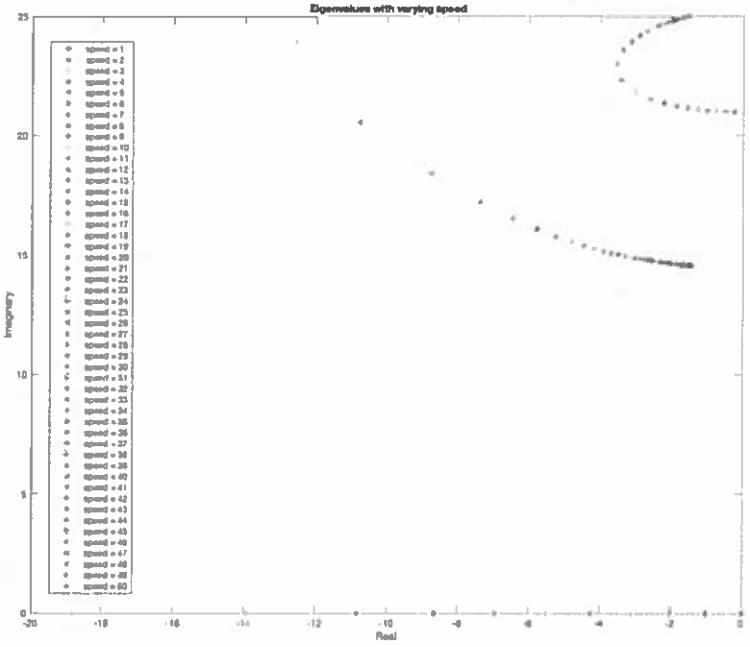
Question 8

- Script

```
clear all
clc
z_initial=0.01;
k=100000;
sigma_rel=0.2;
for i=1:50
    speed=1+1*(i-1);
    [A,B,C,D]=linmod('coursework_model');
    e=eig(A)
    figure(2)
    plot(real(e),imag(e),'*') % Plot real and imaginary parts
    xlabel('Real')
    ylabel('Imaginary')
    title('Eigenvalues with varying speed');
    legendInfo{i}=['speed = ' num2str(speed)];
    i=i+1;
    hold on
end
legend(legendInfo,'southwest')
xlim([-20,0])
ylim([0,25])
hold off
```

Number: 1 Author: Simos Evangelou Date: Indeterminate
Simos Evangelou: So where are the plots? You should plot each against time.

- Plot



Question 9

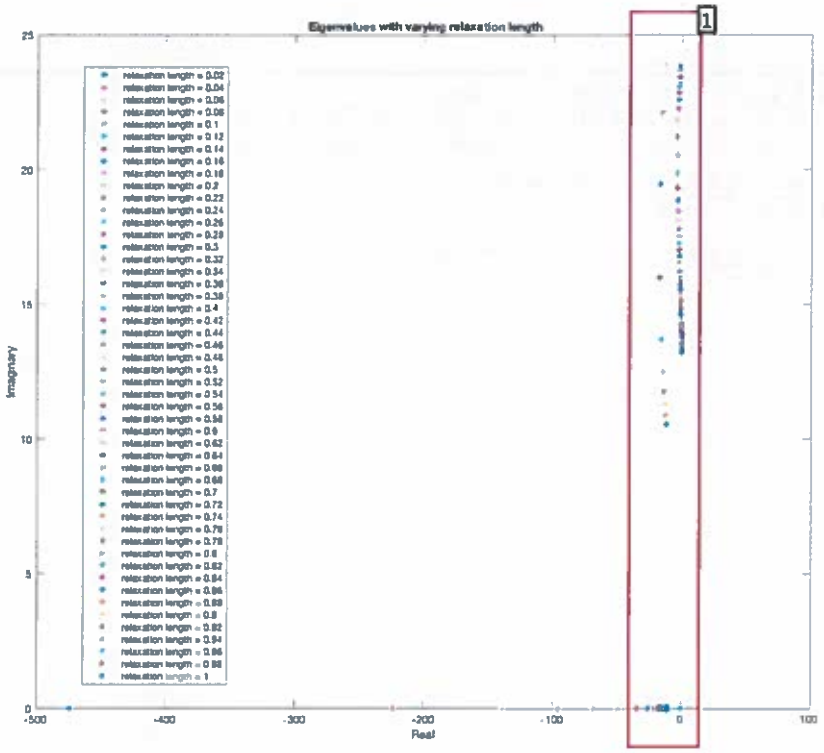
- Script

```
clear all
clc
z_initial=0.01;
k=100000;
speed=10;
for i=1:50
    sigma_rel=0.02+0.02*(i-1);
    [A,B,C,D]=linmod('coursework_model');
    e=eig(A)
    figure(2)
    plot(real(e),imag(e),'*') % Plot real and imaginary parts
    xlabel('Real')
    ylabel('Imaginary')
    title('Eigenvalues with varying relaxation length');
    legendInfo{i}=['relaxation length = ' num2str(sigma_rel)];
    i=i+1;
    hold on
end
legend(legendInfo,'southwest')
ylim([0,25])
hold off
```

- Plot

Page: 6

Number: 1 Author: Simos Evangelou Date: Indeterminate
Simos Evangelou: You did not use axes limits that are big enough to show all the eigenvalues as the speed varies. However, the part that you have show seems correct.


Eigenvalues with varying relaxation length
Imaginary
Real
25
20
15
10
5
0
-500
-400
-300
-200
-100
0
100

1

2

3

Page: 7

Number: 1 Author: Simos Evangelou Date: Indeterminate

Simos Evangelou: You could have zoomed in here to show more clearly how the eigenvalues near the imaginary axis vary, which play the most important role in this case in terms of system stability. Even though not very clear, your root locus is probably correct.

Number: 2 Author: Simos Evangelou Date: Indeterminate

Simos Evangelou: No time responses shown.

Number: 3 Author: Simos Evangelou Date: Indeterminate

Simos Evangelou: Q10 not attempted.